1 Lecture 11 Inference with OLS Short Answer Solutions

2 Question #1

The least squares estimator is

$$b = (X'X)^{-1}X'Y.$$

Under the classical linear regression model assumptions, including normally distributed disturbances, the sampling distribution is

$$b \mid X \sim N\left(\beta, \sigma^2(X'X)^{-1}\right).$$

For confidence intervals and hypothesis testing for β_k, we use the standardized estimate. Explain how the standardized estimate is constructed and describe its sampling distribution.

Solution:

Let $\left[(X'X)^{-1}\right]_{kk}$ denote the kth diagonal element of the matrix inverse. The standardized estimate is

$$\frac{b_k - \beta_k}{\sqrt{\sigma^2\left[(X'X)^{-1}\right]_{kk}}} \sim N(0,1).$$

However, since σ^2 is unobserved, it must be estimated using

$$s^2 = \frac{1}{n-K}\sum_{i=1}^{n} e_i^2.$$

Replacing σ^2 with s^2 changes the sampling distribution of the standardized estimate to a t-distribution with $n-K$ degrees of freedom:

$$\frac{b_k - \beta_k}{\sqrt{s^2\left[(X'X)^{-1}\right]_{kk}}} \sim t_{n-K}.$$

3 Question #2

In the linear regression model:

$$Y = X\beta + \boldsymbol{\varepsilon},$$

define the residual maker for X as

$$M_X = I - X(X'X)^{-1}X'.$$

The OLS estimator b satisfies

$$Y = Xb + e.$$

a. Given that

$$e = M_X\boldsymbol{\varepsilon},$$

show that $e'e = \boldsymbol{\varepsilon}'M_X\boldsymbol{\varepsilon}$ (working through the matrix algebra)

Solution:

Starting with $e = M_X\varepsilon$,

$$e'e = \varepsilon' M_X' M_X \varepsilon.$$

Since M_X is symmetric,

$$\begin{aligned} M_X' M_X &= \left(I - X(X'X)^{-1}X'\right)\left(I - X(X'X)^{-1}X'\right) \\ &= I - 2X(X'X)^{-1}X' + X(X'X)^{-1}X'X(X'X)^{-1}X' \\ &= I - 2X(X'X)^{-1}X' + X(X'X)^{-1}X' \\ &= I - X(X'X)^{-1}X' \\ &= M_X. \end{aligned}$$

Therefore,

$$e'e = \varepsilon' M_X \varepsilon.$$

b. Given that

$$\mathrm{E}(e'e \mid X) = \mathrm{tr}\left(M_X \mathrm{E}(\varepsilon\varepsilon' \mid X)\right),$$

show that under homoskedasticity and non-autocorrelation,

$$\mathrm{E}\left(\frac{e'e}{n-K} \mid X\right) = \sigma^2.$$

(Please, no short cuts. Make all of your steps and simplifications clear)

Solution:

Under homoskedasticity and non-autocorrelation,

$$\mathrm{E}(\varepsilon\varepsilon' \mid X) = \sigma^2 I.$$

Substituting:

$$\begin{aligned} \mathrm{E}(e'e \mid X) &= \mathrm{tr}(M_X \sigma^2 I) \\ &= \sigma^2 \mathrm{tr}(M_X) \\ &= \sigma^2 (n-K). \end{aligned}$$

The final equality follows from

$$\begin{aligned} \mathrm{tr}(M_X) &= \mathrm{tr}(I) - \mathrm{tr}\left(X(X'X)^{-1}X'\right) \\ &= n - \mathrm{tr}\left((X'X)^{-1}X'X\right) \\ &= n - \mathrm{tr}(I_K) \\ &= n - K. \end{aligned}$$

Therefore,

$$\mathrm{E}\left(\frac{e'e}{n-K} \mid X\right) = \sigma^2.$$

4 Question #3

The least squares estimator is

$$b = (X'X)^{-1}X'Y,$$

where X is $n \times K$, Y is $n \times 1$, and b is $K \times 1$. Under the classical assumptions, the variance of the estimator is

$$\text{Var}(b \mid X) = \sigma^2(X'X)^{-1}.$$

Consider the bivariate regression model

$$y_i = \beta_1 + \beta_2 x_i + \varepsilon_i.$$

In this case, X is an $n \times 2$ matrix with a first column of ones and a second column containing the observed values of x_i:

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \qquad Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}.$$

a. Taking σ^2 as given, use matrix algebra to derive the variance of the OLS slope estimator b_2.

Solution:

Here you need to show the matrix algebra leading to

$$\text{Var}(\text{b}_2 \mid \text{X}) = \frac{\sigma^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

b. Using your result from Part (a), identify and explain the three factors that determine the variance of the OLS slope estimator.

Solution:

We can rewrite the denominator using the sample variance of x:

$$s_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

The variance of the OLS slope estimator can then be expressed as

$$\text{Var}(b_2 \mid X) = \frac{\sigma^2}{(n-1)s_x^2}.$$

This highlights three factors that determine the variance of b_2:

1. Disturbance variance: Greater σ^2 increases the variance of b_2.
2. Sample size: More observations reduce the variance of b_2, holding the variation in x fixed.
3. Variation in x: Greater variation in x reduces the variance of b_2, holding sample size fixed.

5 Question #4

Consider the linear regression model:

$$y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \cdots + \beta_K x_{iK} + \varepsilon_i.$$

a. Consider the OLS estimate b_3 of β_3 under the classical assumptions. Write down the formula for the variance of this estimator, taking σ^2 as given.

Solution:

The formula you should have memorized and written down should be of the form

$$\mathrm{Var}(b_3 \mid X) = \frac{\sigma^2}{(1 - R_3^2)\mathrm{SST}_{x_3}}.$$

b. Explain what each of the three components of the formula represents and under what conditions they would improve the precision of the estimator.

Solution:

Explain all three components in detail, making especially clear what the R_3^2 component represents. For example, it comes from a regression of x_3 on all the other regressors and measures the fraction of variation in x_3 explained by those regressors. Explain how this affects the precision of b_3.

See the lecture notes.

6 Question #5

Consider the linear regression model:

$$y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \cdots + \beta_K x_{iK} + \varepsilon_i.$$

a. Explain what multicollinearity is and how it affects the precision of OLS estimates.

Solution:

Multicollinearity describes the degree to which regressors are correlated with one another. If a regressor is highly correlated with the other regressors, its OLS coefficient estimate will be less precise. Recall that

$$\mathrm{Var}(b_k \mid X) = \frac{\sigma^2}{(1 - R_k^2)\mathrm{SST}_{x_k}}.$$

Here, R_k^2 is the fraction of variation in x_k explained by the other regressors. As R_k^2 increases, the denominator decreases, causing $\mathrm{Var}(b_k \mid X)$ to increase.

b. Explain why removing highly correlated regressors may improve precision but can also introduce bias into OLS estimates.

Solution:

Removing highly correlated regressors can reduce R_k^2, increasing the denominator in the variance formula and reducing the variance of b_k.

However, moving these variables from the explained part of the model to the unexplained part may cause a violation of mean independence.

In particular, the nature of multicollinearity is that one variable is highly predictive of another. If one of these variables is moved to the unobserved part of the model, the remaining variables will be highly predictive of the disturbance, which is exactly the situation that creates a violation of mean independence.